

Traceable Secret Sharing: Collusion and Reconstruction Boxes

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In a t -out-of- n secret-sharing scheme, any t shares reconstruct the secret, while fewer reveal nothing. Classical schemes lack accountability: a coalition of $f < t$ parties can collude to build a *reconstruction box* R that, when given $t - f$ additional (honest) shares, outputs the secret without revealing which parties leaked their shares.

Collusion and Reconstruction Boxes

Let $I \subset [n]$, $|I| = f < t$, be the set of corrupted parties. They hardcode their shares $\{(x_i, y_i)\}_{i \in I}$ into a black-box program R , which we call *Reconstruction Box*. By design, R satisfies:

- **Goodness:** For the same polynomial correlation, any input of $t - f$ fresh shares yields the correct secret s^* with probability at least ϵ .
- **Stealth:** R hides I , preventing Alice from knowing the traitors.

Components of traceable (Threshold) Secret Sharing Scheme (TTSS)

A Traceable Threshold Secret Sharing scheme (TTSS) comprises the following PPT algorithms:

- **Share**($1^\lambda, s, n, t, \rho$): Distributes secret s into shares $\{s_i\}$, and outputs tracing keys $\{tk_i\}$ and verification keys $\{vk_i\}$, using correlation string ρ .
- **Rec**($\{s_{i_1}, \dots, s_{i_t}\}$): Reconstructs secret s via interpolation; rejects inconsistent sets.
- **Verify**(s_i, vk_i): Checks individual share consistency against global commitments or keys; ensures correctness before reconstruction.
- **Trace**($R, \{tk_i\}$): On failure or black-box access to a reconstruction box R , identifies a subset of traitors by querying R and analyzing responses with $\{tk_i\}$.

Security Properties of TTSS

- **Secrecy:** Any coalition of fewer than t parties, including the tracer with access to tracing keys, learns no information about the secret beyond a negligible advantage.
- **Traceability:** Any coalition of $f < t$ parties that crafts a reconstruction box R can be algorithmically traced: The *Trace* procedure recovers at least one guilty index from black-box access to R .

- **Non-Imputability:** Even a malicious tracer, given all shares and tracing/verification keys, cannot generate a false proof implicating an honest party: Verification of any such proof will fail except with negligible probability.

Traceability via Reconstruction Boxes (Trace Procedure)

Assume $f = t - 1$ parties are corrupted and their shares $\{(x_i, y_i)\}_{i \in I}$ are hardcoded into R . The *Trace* algorithm, given only black-box access to R , recovers at least one index in I :

1. Query R on a random fresh share (x^*, y^*) to obtain

$$s = R(x^*, y^*) = \sum_{i=1}^f \ell_i(x^*, \dots) y_i + \ell_f(x^*, \dots) y^*.$$

2. Query R on the perturbed share $(x^*, y^* + 1)$ to obtain

$$s' = R(x^*, y^* + 1) = \sum_{i=1}^f \ell_i y_i + \ell_f (y^* + 1).$$

3. Compute the difference

$$\delta = s' - s = \ell_f(x^*, x_1, \dots, x_f),$$

which directly yields the Lagrange coefficient at x^* for the unknown share. This *fingerprint* embeds information about the traitors' x_i values.

4. Define the tracing polynomial

$$h(X) = \prod_{k=1}^f \frac{x_k - X}{x_k},$$

so that each evaluation satisfies

$$h(x^*) = \frac{1}{\ell_f(x^*, x_1, \dots, x_f)} = \delta^{-1}.$$

5. Repeat for t distinct points x^* , interpolate the degree- f polynomial $h(X)$, and factor it. The roots $\{x_1, \dots, x_f\}$ uniquely identify the colluding parties.

Shamir-based Traceable Secret Sharing

Share: Dealer picks random polynomial $q(X) = s + \sum_{j=1}^{t-1} a_j X^j$ and, for each i , samples random x_i , sets $y_i = q(x_i)$, and gives party i the tuple (x_i, y_i) as share, tracing key and verification key.

Reconstruct: Standard Lagrange interpolation on any t shares.

Trace: As above, the interpolation of $h(X)$ yields corrupted x_i values.

Conclusion

Traceable secret sharing preserves confidentiality while deterring collusion: any reconstruction box built by $f < t$ parties can be algorithmically traced back to at least one misbehaving server, enabling accountability in distributed storage and threshold cryptosystems.